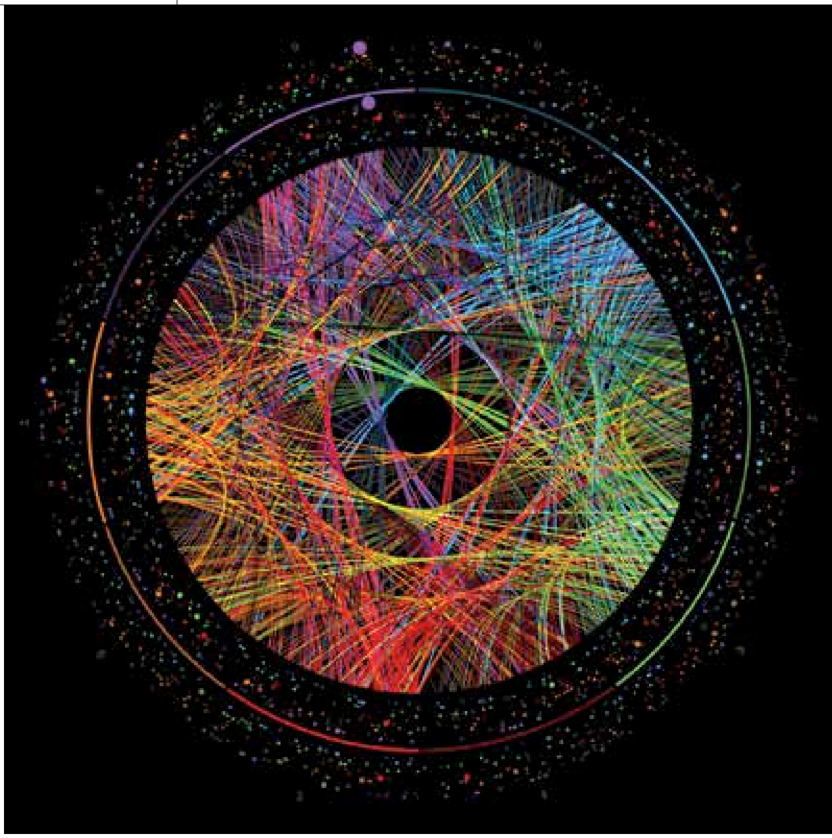
Solving Problems at Lightspeed
Researchers design a new method to calculate properties of nuclei in one hour, rather than 20 years. **http://dgit.in/mar21-25**

Precise Orientation
Engineers developed a technique to precisely orient microscopic devices from folded DNA molecules. **http://dgit.in/mar21-26**



Credit: Martin Krzywinski

THE RAMANUJAN MACHINE

The modern use of Artificial Intelligence to generate mathematical conjectures

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The Ramanujan Machine is a concept developed by scientists from Technion (Israel Institute of Technology). It basically leverages algorithms that perform specific functions for deriving new mathematical identities for fundamental constants such as e, πφ and the golden ratio φ.

OVERVIEW

Throughout the history of mathematics, discoveries and the proving of conjectures for such fundamental constants has been scarce and sporadic. For the approach to these discoveries, mathematicians normally suggest a relationship or a formula for a fundamental constant, and then they try to prove it rigorously. However, the Ramanujan Machine reverses this traditional approach – we first feed in the value of a fundamental constant, and then the algorithm gives out a result. These results are likely to be true, hence, mathematicians can further take those results to formally prove them.This newfound approach to mathematics has created numerous breakthroughs already, and all its proposed conjectures are available online for the global mathematics community to help prove. It has already discovered many known, as well as previously unknown fraction representations of the fundamental constants.

The algorithm follows the way Ramanujan worked during his life (1887 to 1920). He was a child prodigy who managed to bring many mathematical conjectures, and proofs to unsolved equations at the time. He often came up with formulae or identities without any formal training and left the trained mathematicians to formally prove his suggested conjectures.

ITS DEVELOPMENT

Throughout history, simple formulas or identities for fundamental constants have symbolised mathematical beauty. A very well-known example includes Euler's identity $e^{i\pi}+1=0$. Discovering these identities are considered to be a feat of human ingenuity due to the difficulty involved in this process. What's even harder, is to suggest such identities (a conjecture) and to formally prove them with mathematical rigour. It was impossible to attempt automated theorem proving algorithms two decades ago.

Recently, Google developed an AI that automatically proves theorems, in fact it has already proven over 1200 mathematical theorems. This alone is a gargantuan achievement, for any human mathematician, but for the case of computers, it is now the benchmark for our expectations. Hence the use of AI for discovering new conjectures is only the next logical step in this entire process. This is where the Ramanujan Machine comes into play. It aims to automate the process of conjecture generation through new conjectures for fundamental

Srinivasa Ramanujan, child prodigy mathematician